MIL-C-3098/121B
8 March 1983
SUPERSEDING
MIL-C-3098/121A(EL)
23 June 1975

MILITARY SPECIFICATION SHEET

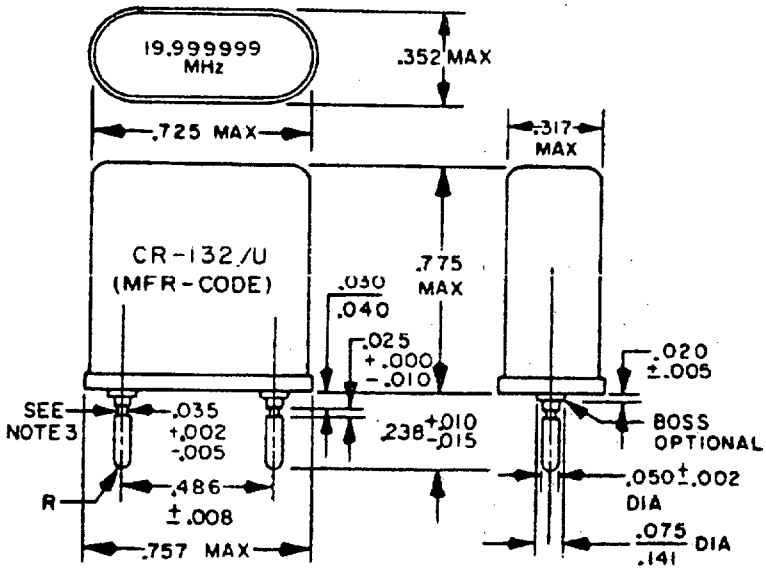
CRYSTAL UNIT, QUARTZ, CR-132/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

(B) INACTIVE FOR NEW DESIGN

Pertinent characteristics: 0.8 to 20 MHz; fundamental; controlled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR-132/U.

(B) denotes changes

FSC 5955

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:56Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (MIL-H-10056/2)
Dimensions, marking, and configuration	See figure 1
Frequency range	0.8 to 20.0 MHz, inclusive
Frequency tolerance	±20 ppm
Equivalent resistance	See table III
Antiresonance, load capacitance	30 ±0.5 pF
Mode of oscillation	Fundamental
Operable temperature range	-55°C to +80°C, inclusive
Operating	80°C to 90°C, inclusive
Test set and calibration values	See table II
Rated drive level	See table II
Capacitance, shunt	7 pF maximum
Shock (specified pulse):	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Frequency stability	±5 ppm
Temperature run:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent

1/ Method 201 of MIL-STD-202.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	0.8 to 1.5	0	1/	----	5.0 ±1.0 2/
	1.5+ to 3.4	50	10	----	5.0 ±1.0
	3.4+ to 5.1	22	15	----	5.0 ±1.0
	5.1+ to 10	13	20	----	5.0 ±1.0
	10+ to 15	11	15	----	2.5 ±0.5
TS-683/TSM	15+ to 20	10	--	0.16	2.5 +0.5

1/ Set crystal current control at extreme counterclock (minimum) position.

2/ The violation of $P=I^2R$ is intentional; at the higher resistance of the crystal unit, the empirical power dissipation will be as rated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:56Z
Check the source to verify that this is the current version before use.

TABLE III. Resistance resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.8 to 0.85	630
0.85+ to 0.9	600
0.9+ to 1	580
1+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.5	410
1.5+ to 1.62	380
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2	290
2+ to 2.12	270
2.12+ to 2.25	250
2.25+ to 2.6	200
2.6+ to 3	150
3+ to 3.4	110
3.4+ to 3.75	90
3.75+ to 4	75
4+ to 5	60
5+ to 7	35
7+ to 10	24
10+ to 15	22
15+ to 20	20

(B) QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTES:

1. The socket designed to mate with crystal holder HC-6/U is Part No. M12883/04-01 (see MIL-S-12283/4).
   For Navy use, other sockets listed in MIL-STD-242 may be used.

2. NATO preferred type NXT 12/B.

3. This crystal unit is identical to the CR-36A/U, except for a load capacitance rating of 30 pF ±0.5 pF. This crystal unit is not interchangeable with the CR-36A/U.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0617-10)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:56Z
Check the source to verify that this is the current version before use.